Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **Python/C++ tooling**, **simulation and data systems**, automation, testing, and **reliable engineering software** across fast-moving production environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Reduced dashboard latency from 1.2s to 350ms through **React interfaces, SQL indexing, and query optimization**, supporting 500+ users with responsive, maintainable engineering workflows across isolated tenant environments.
- Achieved 99%+ reliability across 5K+ monthly jobs through **validation, reconciliation, and failure handling**, cloud storage, CI/CD releases, and production monitoring for dependable data-intensive software workflows at scale.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, secure ingestion, AWS RDS, and CI/CD workflows.
- Processed 1K+ files per batch using **concurrent Python services**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during sustained production traffic and model execution reliably.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **quantitative evaluation pipeline** over 100K+ autonomous-driving logs, combining offline reinforcement learning with reproducible experiment workflows for repeatable analysis.
- Improved closed-loop planning stability by 15% through a **spatial-temporal model** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across simulation scenarios reliably at scale.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **maintainable microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments under sustained traffic.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform**, designing reusable interfaces for allocation, permissions, billing, and notifications with real-time updates across distributed operational locations reliably.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing JWT access controls, WebSocket updates, Azure monitoring, automated tests, and Docker/Jenkins deployments across eight modular service domains.

Skills

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, Bash
Systems: Data Structures, Algorithms, Object-Oriented Programming, Multithreading, Linux, Simulation
Applications: React, Node.js, FastAPI, REST APIs, WebSockets, Microservices, JSON, Caching
Data: PostgreSQL, MySQL, MongoDB, Data Modeling, Schema Design, Query Optimization, Data Validation, Batch Processing
Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD
Quality: Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Debugging, Monitoring, Error Handling, Fault Isolation, Performance Optimization, Production Reliability, Technical Documentation

Projects

Telemetry Test Workbench - Design Concept
- Designed a **Python/C++ telemetry workbench concept** for sensor streaming, validation, synchronized visualization, anomaly detection, replayable test sessions, and structured exports supporting rapid diagnosis across engineering workflows reliably.

Manufacturing Workflow Automation - Design Concept
- Designed a **TypeScript/React operations tool concept** connecting work orders, technician inputs, validation rules, status events, and audit histories through APIs, enabling clear human and agent-driven workflows for daily use.

CPU-scheduling Simulator | Live
- Implemented **FCFS, SJF, SRT, Round Robin, priority, and MLQ scheduling** with React controls, EJ-2 visualization, and dynamic turnaround and waiting-time calculations across 1K+ simulated process combinations reliably.